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NanoTeacher-YOLO26: Foundation-to-Edge Self-Supervised Distillation for Lightweight Potato Disease Detection

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Abstract

Lightweight potato disease detectors are promising for field deployment, but their compact backbones often lose the fine spatial evidence required to distinguish early blight, late blight, and healthy leaves. This paper presents NanoTeacher-YOLO26, a foundation-to-edge self-supervised distillation framework that transfers teacher-side visual priors into a deployable YOLO26n student without increasing inference cost. A stronger YOLO26s teacher is used only during training to provide structured disease-aware supervision, including confidence-weighted feature geometry, disease-evidence attention, and boundary-gradient priors. These signals are integrated into a Disease-Aware Distillation Loss, enabling the nano student to inherit the teacher's localization behavior and spatial response patterns while preserving its original model footprint. Experiments are conducted on a YOLO-format potato leaf disease benchmark derived from the public PlantVillage potato subset. Results show that NanoTeacher-YOLO26 reaches the saturated mAP regime of the benchmark and improves over the same-footprint YOLO26n baseline in precision and recall. Detection overlays, attention maps, teacher-student alignment, confusion analysis, and training curves further indicate that the distilled student learns disease-relevant spatial evidence rather than relying only on class-level shortcuts. The study provides a reproducible route for converting teacher-side perception into edge-ready agricultural detectors.

Keywords Edge intelligence · Knowledge distillation · Potato disease detection · YOLO26n

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Introduction

Potato production is persistently threatened by foliar diseases whose symptoms appear as local texture, color, shape, and boundary changes on leaves. Early blight and late blight are visually distinct in advanced stages, but their field-facing recognition remains difficult because symptoms may be partial, weakly contrasted, or distributed across several small regions. Manual scouting is labor intensive, while a pure image classifier provides only an image-level decision and gives limited evidence about where the model finds disease cues. For practical precision agriculture, a detector should be accurate, fast, visually auditable, and small enough to be deployed on low-resource devices.

Deep learning has become the dominant paradigm for plant disease recognition. PlantVillage and related public datasets enabled large-scale training of convolutional networks for crop-disease diagnosis (Mohanty et al. 2016; Arun Pandian and Geetharamani 2019). In potato research, recent studies have reported strong classification performance through optimized CNNs, CNN-LSTM models, attention modules, feature-selection strategies, and transformer-style designs (Elshewey et al. 2025; El-Kenawy et al. 2025; Alzakari et al. 2025; Bajpai et al. 2025; Tiwari and Rajput 2025; Kaur et al. 2025; Ozkan 2026). These works confirm that deep networks can model potato leaf symptoms, but many of them are built as classifiers. A classifier can answer which disease is present, yet it does not natively provide bounding boxes, detector confidence, or localized visual evidence for downstream screening systems.

YOLO-family detectors are suitable for this setting because they perform localization and classification in a single pass (Redmon et al. 2016; Jocher et al. 2026). Nevertheless, simply choosing a nano detector creates a tension between deployment efficiency and representational richness. A small YOLO model may have enough capacity for controlled leaf images, but the model can still underuse subtle lesion boundaries and symptom-specific spatial patterns. Increasing the backbone scale is a direct remedy, but that solution moves in the opposite direction of edge deployment.

This paper investigates a different path: using a stronger teacher model during training and deploying only a compact student model. The proposed framework, called **NanoTeacher-YOLO26**, uses a YOLO26s teacher to provide structured visual priors to a YOLO26n student. We use the term *foundation-to-edge* in an operational sense: teacher-side perception is used as a training-time source of structured supervision, while the final detector remains an edge-sized model. The present implementation does not claim that YOLO26s is itself a vision foundation model comparable to Segment Anything or Grounding DINO. Instead, the framework is designed to be compatible with such teachers in future extensions, whereas the current study deliberately adopts a same-family YOLO26s teacher to keep the experimental protocol reproducible, lightweight, and directly comparable with the YOLO26n student. Similarly, *self-supervised* refers to the teacher-generated feature, attention, and boundary targets used during student training; these auxiliary spatial targets do not require additional lesion-level annotations, although the teacher detector is trained from the available leaf-level labels.

The central technical component is a **Disease-Aware Distillation Loss**. Unlike conventional knowledge distillation that transfers only logits or final predictions

(Hinton et al. 2015), the proposed loss transfers three kinds of spatial supervision: confidence-weighted feature geometry, normalized disease-evidence attention, and attention-boundary gradients. The first term encourages the student to match the teacher where the teacher has high detection confidence. The second term aligns spatial attention maps extracted from teacher and student feature tensors. The third term transfers horizontal and vertical attention gradients so that disease boundaries and lesion-like transitions are not erased by the nano backbone. During inference, the teacher and all projector modules are removed; only the YOLO26n student is retained.

The main contributions are summarized as follows:

1. A foundation-to-edge training framework is developed for lightweight potato disease detection, in which a strong teacher detector supervises a deployable YOLO26n student.
2. A Disease-Aware Distillation Loss is proposed to jointly transfer confidence-weighted features, disease-evidence attention, and boundary-gradient priors.
3. A reproducible YOLO-format benchmark is constructed from the public PlantVillage potato subset, including early blight, healthy leaf, and late blight categories.
4. A complete visualization package is provided, including detection overlays, attention heatmaps, teacher-student alignment maps, confusion analysis, training curves, and model-footprint comparisons.

Related Work

Potato Disease Recognition

The development of public plant disease datasets has greatly accelerated automatic crop disease recognition. Mohanty et al. (2016) demonstrated the feasibility of deep learning for image-based plant disease diagnosis using a large public leaf dataset. The dataset release by Arun Pandian and Geetharamani (2019) and the nine-layer CNN study by Geetharamani and Arun Pandian (2019) further supported reproducible plant leaf disease experiments across many crop categories, including potato early blight, late blight, and healthy leaves.

Potato Research has recently published a concentrated body of work on potato disease intelligence. Elshewey et al. (2025) optimized potato blight detection through waterwheel plant and sine cosine algorithms. El-Kenawy et al. (2025) used a hybrid metaheuristic strategy to optimize CNN-based potato disease classification. Alzakari et al. (2025) introduced a CNN-LSTM model for early detection of potato disease, while Bajpai et al. (2025) investigated attention and squeeze-and-excitation mechanisms for potato leaf disease recognition. Kaur et al. (2025) proposed PotConvNet as an automated CNN framework, and Tiwari and Rajput (2025) studied super-resolution and multi-path attention transformers. Ozkan (2026) compared deep features with traditional classifiers under both controlled and field conditions. Gülmez (2025) provided a broad review of CNN-based potato disease detection strategies.

These studies show that the potato disease domain has moved beyond simple baseline classification. However, classification-oriented systems still have limitations for

localization, model auditing, and integration with robotic scouting. Hulsman et al. (2025) used activation maps for potato blackleg localization and emphasized the value of explainable visual evidence. The present work follows that motivation but trains an explicit object detector and uses teacher-student distillation to keep the deployed model compact.

Teacher-Student Distillation

Knowledge distillation transfers information from a high-capacity teacher model to a smaller student model (Hinton et al. 2015). Early distillation methods focused mainly on softened class probabilities, but dense prediction tasks require richer forms of transfer. For detection, the student must learn not only class confidence but also multi-scale feature layout, localization behavior, and spatial evidence. A naive global feature loss can waste capacity on background regions, especially in agricultural images where disease evidence is sparse or localized.

The proposed distillation strategy is therefore disease-aware. Teacher confidence is used to weight feature regression so that the student spends more learning capacity on regions likely to contain disease-relevant evidence. Attention alignment transfers the teacher's spatial response pattern, while boundary-gradient alignment preserves transitions around leaf edges, lesion-like spots, and symptom contours. This design is related in spirit to explainability methods such as Grad-CAM, which visualize discriminative regions in CNNs (Selvaraju et al. 2017), but here attention maps are not only post-hoc explanations; they also become training signals for a compact detector.

Foundation Models and Edge Agricultural Vision

Vision foundation models have changed how visual supervision can be produced. Segment Anything introduced a promptable segmentation model with strong zero-shot behavior (Kirillov et al. 2023), and Grounding DINO connected language prompts with open-set object detection (Liu et al. 2023). Such models are promising teacher sources for agriculture because they can generate masks, boxes, and semantic proposals with limited manual annotation. Their inference cost, however, is often too high for small agricultural edge devices.

NanoTeacher-YOLO26 adopts the same training philosophy while using a lightweight teacher instance for reproducibility. The teacher model is allowed to be stronger than the deployed student, but the final detector remains a nano YOLO26n. This separation is important: the heavy side of perception is used during training, while the deployed side remains fast. The result is a practical compromise between foundation-style supervision and edge-constrained inference. We therefore interpret foundation-to-edge as a transfer paradigm rather than as a claim that the implemented teacher is a general-purpose foundation model.

Methods

Task Definition

Let $\mathcal{D} = \{(\mathbf{I}_i, \mathcal{Y}_i)\}_{i=1}^{N}$ denote the potato disease detection dataset, where $\mathbf{I}_i \in \mathbb{R}^{3\times H\times W}$ is an input image and $\mathcal{Y}_i$ is the corresponding detection annotation set. For image $\mathbf{I}_i$, the annotation set is written as

$$\mathcal{Y}_i = \left\{(\mathbf{b}_{ij}, y_{ij})\right\}_{j=1}^{M_i}, \tag{1}$$

where $\mathbf{b}_{ij} = (x_{ij}, y_{ij}, w_{ij}, h_{ij})$ is a normalized bounding box in center-format coordinates, $y_{ij} \in \mathcal{C}$ is the category label, and $\mathcal{C}$ contains early blight, healthy leaf, and late blight. Although the benchmark used in this study contains one dominant leaf-level object per image, the formulation is kept general to match the standard object-detection setting.

Given an input image $\mathbf{I}$, a detector parameterized by θ predicts a set of detection hypotheses:

$$f_\theta(\mathbf{I}) = \left\{(\hat{\mathbf{b}}_k, \hat{\mathbf{p}}_k, \hat{s}_k)\right\}_{k=1}^{K}, \tag{2}$$

where $\hat{\mathbf{b}}_k$ is a predicted bounding box, $\hat{\mathbf{p}}_k \in [0, 1]^{|\mathcal{C}|}$ is the class-probability vector, and $\hat{s}_k$ is the confidence score. The standard YOLO detection objective can be written in a compact form as

$$\mathcal{L}_{\text{det}} = \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{dfl}}\mathcal{L}_{\text{dfl}}, \tag{3}$$

where $\mathcal{L}_{\text{box}}$ supervises localization, $\mathcal{L}_{\text{cls}}$ supervises disease-category prediction, and $\mathcal{L}_{\text{dfl}}$ is the distributional box-regression loss inherited from the YOLO26 training objective.

NanoTeacher-YOLO26 does not modify the inference-side architecture of the student detector. Instead, it introduces a training-only teacher-student pathway. The teacher provides structured intermediate supervision, while the final deployed model remains the original YOLO26n student. Therefore, the proposed method improves the optimization and representation of the nano detector without increasing the deployment cost.

Because the benchmark is converted from classification-oriented potato leaf images, the bounding box describes the dominant leaf region rather than a manually delineated lesion. Thus, the task should be interpreted as leaf-level disease detection with explicit localization, rather than pixel-level lesion segmentation. This distinction is important for interpreting the reported detection metrics: a correct prediction means that the model identifies the diseased or healthy leaf object and assigns the correct class, not that it precisely outlines every lesion spot.

Overall Framework

Figure 1 illustrates the NanoTeacher-YOLO26 framework. The training procedure contains two stages. In the first stage, a stronger YOLO26s teacher detector f_T is trained with the standard detection objective:

$$\theta_T^* = \arg\min_{\theta_T} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_{\text{det}} \left(f_T(\mathbf{I}_i; \theta_T), \mathcal{Y}_i \right). \tag{4}$$

After training, the teacher parameters are frozen. In the second stage, a compact YOLO26n student detector f_S is optimized by both the supervised detection loss and the teacher-guided distillation loss:

$$\theta_S^* = \arg\min_{\theta_S} \frac{1}{N} \sum_{i=1}^{N} \left[\mathcal{L}_{\text{det}} \left(f_S(\mathbf{I}_i; \theta_S), \mathcal{Y}_i \right) + \lambda_d \mathcal{L}_{\text{DAD}} \left(f_S, f_T; \mathbf{I}_i \right) \right], \tag{5}$$

where $\mathcal{L}_{\text{DAD}}$ denotes the Disease-Aware Distillation Loss and λ_d controls the overall strength of teacher-side supervision.

During inference, the teacher network, feature projectors, and distillation losses are removed. The deployed detector is therefore

$$f_{\text{deploy}}(\mathbf{I}) = f_S(\mathbf{I}; \theta_S^*). \tag{6}$$

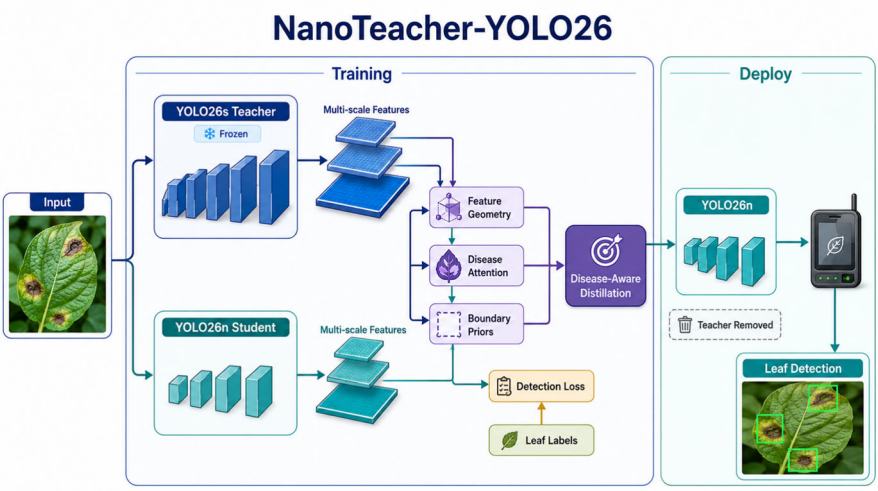


Fig. 1 NanoTeacher-YOLO26 framework. A stronger teacher provides confidence-weighted feature geometry, disease-evidence attention, and boundary priors during training

This design follows a foundation-to-edge principle: a stronger model is allowed to shape the learning dynamics during training, but only the lightweight student is retained for field-oriented deployment.

In implementation, the distillation feature set $\mathcal{S}$ is selected from the multi-scale neck outputs that feed the YOLO Detect head. For YOLO26, the hook-based implementation automatically detects the feature layers {16, 19, 22} and uses the Detect head at layer 23 only to obtain teacher confidence responses. These layers correspond to progressively coarser spatial scales, allowing the student to receive both fine leaf-boundary cues and higher-level disease-category evidence. We selected the Detect-head input layers because they are directly responsible for localization and classification, while earlier backbone layers are more generic and would increase memory use during distillation without necessarily improving edge-relevant detection behavior.

Teacher-Guided Multi-Scale Feature Transfer

Let $\mathcal{S}$ denote the set of selected feature levels used for distillation. At each level $l \in \mathcal{S}$, the teacher and student generate feature tensors

$$\mathbf{F}_l^T \in \mathbb{R}^{C_l^T \times H_l \times W_l}, \qquad \mathbf{F}_l^S \in \mathbb{R}^{C_l^S \times H_l \times W_l}. \tag{7}$$

Because the channel dimensions of the teacher and student may differ, a lightweight projection layer $\phi_l(\cdot)$ is used to align the student feature dimension with the teacher feature dimension:

$$\tilde{\mathbf{F}}_l^S = \phi_l\left(\mathbf{F}_l^S\right), \qquad \tilde{\mathbf{F}}_l^S \in \mathbb{R}^{C_l^T \times H_l \times W_l}. \tag{8}$$

The projection layers are used only during training and are discarded after student optimization.

A direct feature-regression loss treats foreground and background regions equally. However, potato disease evidence is usually spatially sparse, and excessive background matching may waste the limited capacity of the nano student. To make the transfer selective, the teacher confidence is used to construct a spatial weighting map. Let $\mathbf{Z}^T$ denote the teacher-side dense class logits. The confidence map at feature level l is defined as

$$q_l(u, v) = \Pi_l\left(\max_{c \in \mathcal{C}} \sigma\left(\mathbf{Z}_c^T\right)\right)(u, v), \tag{9}$$

where $\sigma(\cdot)$ is the sigmoid function and $\Pi_l(\cdot)$ resizes the teacher confidence response to the spatial resolution $H_l \times W_l$. To stabilize the loss magnitude across batches and feature levels, the confidence map is normalized as

$$\omega_l(u, v) = \frac{q_l(u, v)}{\frac{1}{H_l W_l} \sum_{u=1}^{H_l} \sum_{v=1}^{W_l} q_l(u, v) + \epsilon}. \tag{10}$$

For notational clarity, the channel vector at spatial location (u, v) is denoted as

$$\mathbf{f}_l^T(u, v) = \mathbf{F}_l^T(:, u, v), \qquad \tilde{\mathbf{f}}_l^S(u, v) = \tilde{\mathbf{F}}_l^S(:, u, v). \tag{11}$$

The confidence-weighted feature regression loss is then formulated as

$$\mathcal{L}_{\text{feat}} = \sum_{l \in \mathcal{S}} \frac{1}{C_l^T H_l W_l} \sum_{u=1}^{H_l} \sum_{v=1}^{W_l} \omega_l(u, v) \left\| \tilde{\mathbf{f}}_l^S(u, v) - \mathbf{f}_l^T(u, v) \right\|_2^2. \tag{12}$$

This term encourages the compact student to approximate the teacher's multi-scale feature geometry, while teacher-confident disease regions receive stronger supervision than low-confidence background regions.

Disease-Evidence Attention Alignment

Feature-level regression transfers channel-wise representations, but it does not explicitly constrain the spatial distribution of disease evidence. To transfer teacher-side spatial focus, a normalized attention map is extracted from each feature tensor. For a generic feature tensor $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$, the local evidence energy is defined as

$$E_{\mathbf{F}}(u, v) = \frac{1}{C} \sum_{c=1}^{C} |\mathbf{F}_c(u, v)|. \tag{13}$$

The corresponding normalized spatial attention map is

$$\mathcal{A}(\mathbf{F})(u, v) = \frac{E_{\mathbf{F}}(u, v)}{\frac{1}{HW} \sum_{u=1}^{H} \sum_{v=1}^{W} E_{\mathbf{F}}(u, v) + \epsilon}. \tag{14}$$

This normalization removes scale differences across feature levels and emphasizes the relative spatial concentration of disease-related responses.

The attention maps of the projected student feature and the teacher feature are written as

$$\mathcal{A}_l^S = \mathcal{A}\left(\tilde{\mathbf{F}}_l^S\right), \qquad \mathcal{A}_l^T = \mathcal{A}\left(\mathbf{F}_l^T\right). \tag{15}$$

The Smooth-L_1 penalty used for attention alignment is defined as

$$\rho(a, b) = \begin{cases} \frac{1}{2}(a-b)^2, & \text{if } |a-b| < 1, \\ |a-b| - \frac{1}{2}, & \text{otherwise.} \end{cases} \tag{16}$$

The disease-evidence attention alignment loss is then given by

$$\mathcal{L}_{\text{attn}} = \sum_{l \in \mathcal{S}} \frac{1}{H_l W_l} \sum_{u=1}^{H_l} \sum_{v=1}^{W_l} (1 + \omega_l(u, v))\, \rho\left(\mathcal{A}_l^S(u, v), \mathcal{A}_l^T(u, v)\right). \tag{17}$$

The term $1 + \omega_l(u, v)$ keeps global attention consistency while assigning additional weight to teacher-confident regions. As a result, the student is encouraged not only to match teacher features, but also to reproduce the teacher's spatial evidence layout.

Boundary-Gradient Distillation

Potato disease symptoms often appear as local transitions between healthy and necrotic tissue, irregular lesion contours, or texture discontinuities along diseased regions. A compact detector may over-smooth these transitions because of limited channel capacity. To preserve such structural cues, NanoTeacher-YOLO26 further aligns the spatial gradients of teacher and student attention maps.

For an attention map $\mathcal{A} \in \mathbb{R}^{H \times W}$, the vertical and horizontal finite differences are defined as

$$\Delta_h \mathcal{A}(u, v) = \mathcal{A}(u+1, v) - \mathcal{A}(u, v), \qquad 1 \le u < H, \tag{18}$$

and

$$\Delta_w \mathcal{A}(u, v) = \mathcal{A}(u, v+1) - \mathcal{A}(u, v), \qquad 1 \le v < W. \tag{19}$$

The boundary-gradient distillation loss is formulated as

$$\begin{aligned} \mathcal{L}_{\text{bd}} = & \sum_{l \in \mathcal{S}} \frac{1}{(H_l - 1) W_l} \sum_{u=1}^{H_l - 1} \sum_{v=1}^{W_l} \rho \left(\Delta_h \mathcal{A}_l^S(u, v), \Delta_h \mathcal{A}_l^T(u, v) \right) \\ & + \sum_{l \in \mathcal{S}} \frac{1}{H_l (W_l - 1)} \sum_{u=1}^{H_l} \sum_{v=1}^{W_l - 1} \rho \left(\Delta_w \mathcal{A}_l^S(u, v), \Delta_w \mathcal{A}_l^T(u, v) \right). \end{aligned} \tag{20}$$

This term transfers the teacher's boundary-sensitive response pattern to the student. Compared with raw attention matching, boundary-gradient distillation focuses on how the attention changes across neighboring spatial positions, which is useful for retaining lesion-like transitions and disease-associated local structures.

Disease-Aware Distillation Objective

The final Disease-Aware Distillation Loss combines confidence-weighted feature transfer, disease-evidence attention alignment, and boundary-gradient distillation:

$$\mathcal{L}_{\text{DAD}} = \mathcal{L}_{\text{feat}} + \alpha \mathcal{L}_{\text{attn}} + \beta \mathcal{L}_{\text{bd}}. \tag{21}$$

The complete training objective for the student is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \lambda_d \left(\mathcal{L}_{\text{feat}} + \alpha \mathcal{L}_{\text{attn}} + \beta \mathcal{L}_{\text{bd}} \right). \tag{22}$$

In this study, $\lambda_d = 4.0$, $\alpha = 0.35$, and $\beta = 0.15$. These weights make the feature-level term the dominant distillation signal, while attention alignment and boundary-gradient alignment serve as auxiliary spatial regularizers. They were selected by a validation-oriented heuristic rather than by an exhaustive grid search: the overall distillation multiplier was increased until the distillation loss had a visible optimization effect without overwhelming the supervised YOLO detection loss, and the attention and boundary terms were kept smaller because they regularize spatial layout rather than

Table 1 Component-level interpretation of the Disease-Aware Distillation Loss

Component	Transferred signal	Expected effect if removed
$\mathcal{L}_{\text{feat}}$	Confidence-weighted multi-scale feature geometry from the teacher	The student loses the main teacher-side structural signal and relies mainly on hard labels, reducing spatial consistency under the same nano capacity
$\mathcal{L}_{\text{attn}}$	Normalized disease-evidence attention over leaf regions	The student can still learn boxes, but its activation may become less concentrated on symptom-relevant tissue and more sensitive to background or class-level shortcuts
$\mathcal{L}_{\text{bd}}$	Horizontal and vertical gradients of attention maps	The student may over-smooth local transitions, weakening boundary-like responses around lesions, leaf edges, and symptom texture changes

replace the feature geometry term. Settings with an excessively large distillation multiplier caused the student to track teacher responses too aggressively, whereas very small values behaved similarly to the plain YOLO26n baseline. The final setting was therefore chosen to preserve stable detection training while injecting teacher-side spatial guidance. The teacher is frozen throughout student training, and gradients are back-propagated only through the student detector and the temporary projection layers (Table 1).

This component-level analysis explains why the three terms are complementary. The feature term is the primary capacity-transfer channel, the attention term controls where the compact detector looks, and the boundary term controls how sharply the spatial evidence changes. Together, they regularize the student toward teacher-like localization behavior without adding inference-time parameters.

Algorithm 1 summarizes the training procedure of NanoTeacher-YOLO26.

Algorithm 1 Training procedure of NanoTeacher-YOLO26

Require: Potato detection dataset $\mathcal{D}$, teacher detector f_T, student detector f_S
Train YOLO26s teacher f_T on $\mathcal{D}$ using $\mathcal{L}_{\text{det}}$
Freeze the teacher parameters θ_T^*
Initialize YOLO26n student f_S with pretrained weights
Initialize feature projection layers $\{\phi_l\}_{l \in \mathcal{S}}$
for each mini-batch **do**
  Apply YOLO data augmentations and resize images to 416 pixels
  Forward images through the frozen teacher and trainable student
  Capture multi-scale teacher and student feature tensors with forward hooks
  Compute standard YOLO detection loss $\mathcal{L}_{\text{det}}$
  Compute confidence-weighted feature loss $\mathcal{L}_{\text{feat}}$
  Compute disease-evidence attention loss $\mathcal{L}_{\text{attn}}$
  Compute boundary-gradient loss $\mathcal{L}_{\text{bd}}$
  Compute $\mathcal{L}_{\text{total}}$
  Update the student parameters and projection layers
end for
Discard the teacher and projection layers
Deploy only the YOLO26n student detector

Inference

After training, NanoTeacher-YOLO26 uses only the distilled YOLO26n student for inference. Given a test image $\mathbf{I}$, the prediction is obtained as

$$\left\{(\hat{\mathbf{b}}_k, \hat{\mathbf{p}}_k, \hat{s}_k)\right\}_{k=1}^{K} = f_S(\mathbf{I}; \theta_S^*). \tag{23}$$

No teacher model, projection layer, attention branch, or boundary-gradient computation is used at this stage. Therefore, the inference-side parameter count and checkpoint size are identical to those of the original YOLO26n student. The proposed method can be understood as a training-time regularization strategy that transfers teacher-side spatial knowledge into a compact detector while preserving edge deployment efficiency.

Experimental Setup

Dataset

The dataset is derived from the public PlantVillage leaf image release distributed through Mendeley Data (Arun Pandian and Geetharamani 2019). The potato subset contains three categories: potato early blight, potato healthy, and potato late blight. Because the source images are classification samples rather than object-detection annotations, we converted them into a YOLO-format benchmark using one leaf-level box per image. Specifically, the class label was inherited from the original folder name, and each image was assigned a normalized bounding box $(x, y, w, h) = (0.5, 0.5, 0.98, 0.98)$ to cover the dominant centered leaf while leaving a small border margin. The same conversion rule was applied to all classes and all data splits, producing 2152 images and 2152 object instances. A fixed random seed was used to create the class-aware training, validation, and testing splits reported in Table 2. This construction makes the benchmark reproducible, but it also means that the boxes are leaf-level regions rather than lesion-level boxes. Consequently, the benchmark evaluates whether a compact detector can localize the leaf object and classify its disease status, not whether it can segment or count individual lesions (Fig. 2).

Implementation Details

All models were trained in the local Python 3.8 environment using the Ultralytics YOLO codebase. Images were resized to 416 pixels. AdamW was used with an initial

Table 2 Dataset split and class distribution

Split	Images	Instances	Early blight	Healthy	Late blight
Training	1721	1721	800	121	800
Validation	215	215	100	15	100
Testing	216	216	100	16	100

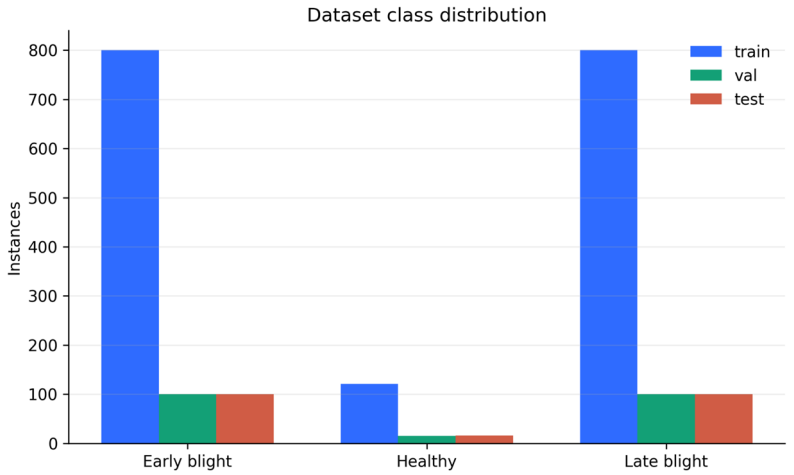


Fig. 2 Class distribution of the YOLO-format potato disease benchmark. Healthy leaves are fewer than early blight and late blight samples, making recall on the healthy class more sensitive to individual images

learning rate of 0.0015, cosine scheduling, batch size 16 for nano models, and batch size 12 for the YOLO26s teacher. The YOLO26s teacher was trained for 14 epochs. YOLO26n, YOLO11n, YOLOv8n, and YOLOv5n-u baselines were trained for 12 epochs under the same resolution and data split. NanoTeacher-YOLO26 was trained for 18 epochs with the frozen YOLO26s teacher and $\lambda_d = 4.0$. For the distilled student, the internal loss terms are denoted consistently as $\mathcal{L}_{\text{feat}}$, $\mathcal{L}_{\text{attn}}$, $\mathcal{L}_{\text{bd}}$, $\mathcal{L}_{\text{DAD}}$, and $\mathcal{L}_{\text{total}}$ in both the equations and Algorithm 1.

The training strategy was intentionally short because the target setting is fast, reproducible edge-model development rather than exhaustive large-scale pretraining. Standard YOLO augmentations were retained, including horizontal flip, color jitter, translation, scaling, and controlled mosaic augmentation. The best checkpoint was selected according to validation mAP and then evaluated on the held-out test split.

Evaluation Metrics

The experiments report precision, recall, mAP_{50}, and $\text{mAP}_{50:95}$. Precision measures the reliability of positive predictions, recall measures the ability to recover target leaves, mAP_{50} evaluates average precision at an IoU threshold of 0.50, and $\text{mAP}_{50:95}$ averages over thresholds from 0.50 to 0.95. Parameter count and checkpoint size are also reported because the main objective is lightweight deployment.

Results and Discussion

Overall Performance

Table 3 reports test-set results for YOLO-family nano baselines and NanoTeacher-YOLO26. The benchmark is close to saturation for all modern nano detectors, so

Table 3 Controlled comparison of nano YOLO detectors on the held-out test split. Values are measured from the completed local experiments

Model	Params (M)	Weight (MB)	Precision	Recall	mAP_{50}	$mAP_{50:95}$
YOLOv5n-u	2.509	5.00	0.9978	1.0000	0.9950	0.9950
YOLOv8n	3.011	5.94	0.9920	0.9907	0.9950	0.9950
YOLO11n	2.590	5.20	0.9983	1.0000	0.9950	0.9950
YOLO26n	2.505	5.12	0.9900	0.9901	0.9950	0.9950
NanoTeacher-YOLO26	2.505	5.12	0.9979	0.9948	0.9950	0.9950

mAP values should be interpreted with care and should not be read as evidence of a large numerical detection gap. The most diagnostic comparison is the same-footprint pair: plain YOLO26n and the distilled YOLO26n student. NanoTeacher-YOLO26 keeps the same 2.505 million parameters and the same 5.12 MB checkpoint size as YOLO26n, while improving precision from 0.9900 to 0.9979 and recall from 0.9901 to 0.9948. Thus, the main empirical message is not that the method substantially raises an already saturated mAP ceiling, but that teacher-guided spatial distillation improves precision–recall behavior and visual alignment without increasing the deployed model size.

The table and Figs. 3 and 4 also show why the task should not be treated as a simple leaderboard. Several nano models reach the same rounded mAP ceiling, and some baselines achieve marginally higher recall because the test set contains only one object per image and the background is controlled. Under such saturation, the essential contribution is not a large mAP gap but the transfer of teacher-side spatial behavior into the same YOLO26n deployment envelope. This interpretation is supported by visual evidence in later subsections.

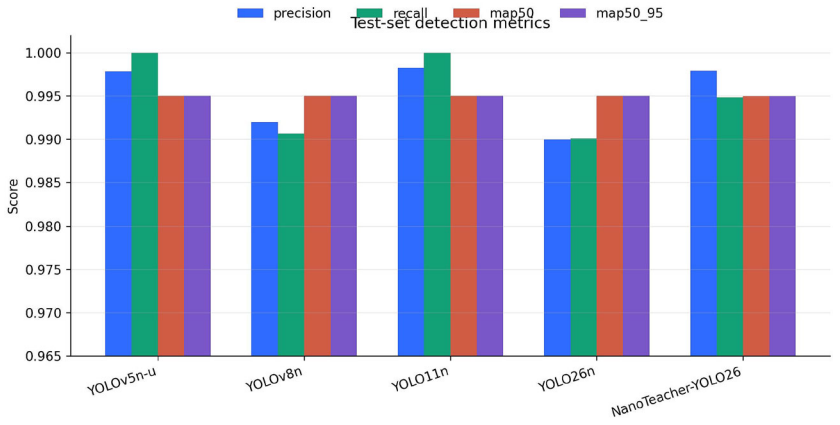


Fig. 3 Test metric comparison on the held-out split. Because the benchmark is saturated, precision and recall differences are more informative than the identical rounded mAP values

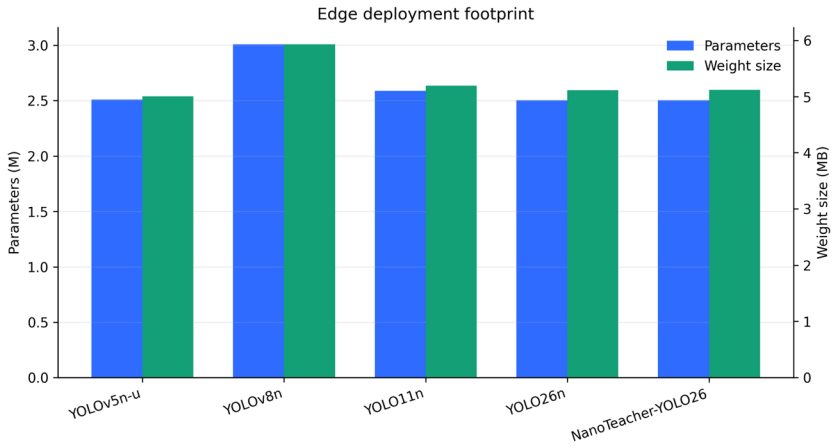


Fig. 4 Model footprint comparison. NanoTeacher-YOLO26 retains the same parameter count and checkpoint size as the plain YOLO26n student because the teacher and projection layers are removed after training

Class-Level Performance

Table 4 gives class-wise performance for NanoTeacher-YOLO26. Healthy leaves are the minority class, with only 16 test samples, yet the model preserves high precision and recall. Late blight reaches perfect recall, while early blight remains close to the ceiling. This behavior indicates that the teacher-student training does not simply overfit the dominant disease classes; it also retains a separable representation for the healthy category. One reason is that the distillation signal is spatial rather than purely frequency-driven: each image contributes teacher confidence, attention layout, and boundary-gradient supervision over its leaf region, so healthy leaves still provide structured teacher targets even though their sample count is smaller. In addition, the supervised classification loss remains active, and the fixed class-aware split ensures that healthy samples are present in training, validation, and testing. Nevertheless, because the healthy test subset is small, these class-wise numbers should be interpreted as supportive evidence rather than as a definitive field-level robustness guarantee.

Table 4 Class-level test performance of NanoTeacher-YOLO26

Class	Precision	Recall	mAP_{50}	$mAP_{50:95}$
Potato early blight	1.0000	0.9899	0.9949	0.9949
Potato healthy	1.0000	0.9946	0.9950	0.9950
Potato late blight	0.9938	1.0000	0.9950	0.9950
All	0.9979	0.9948	0.9950	0.9950

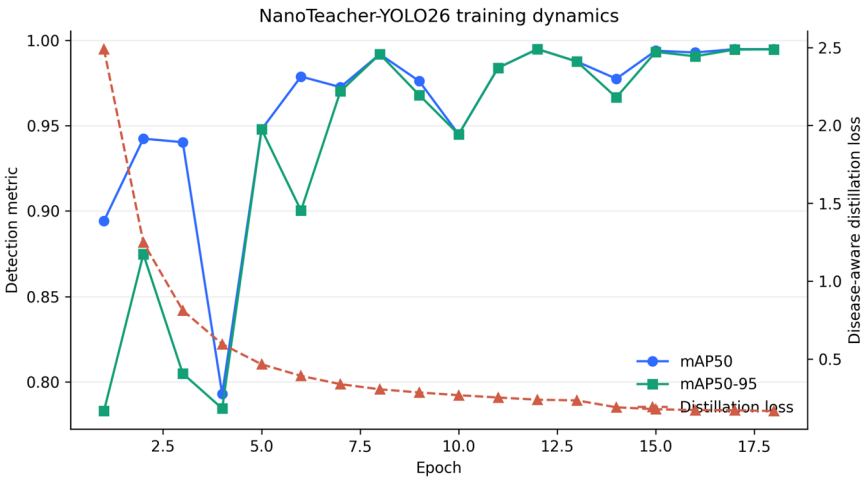


Fig. 5 Training curves for NanoTeacher-YOLO26. The student converges quickly and maintains stable high validation performance while the distillation loss decreases

Training Dynamics and Confusion Analysis

Figures 5 and 6 show the training dynamics and confusion matrix. The mAP curves quickly enter a stable high-accuracy region, while the distillation loss decreases as the student aligns with the teacher. The confusion matrix confirms that the remaining errors are small and that the three classes are well separated under the leaf-level

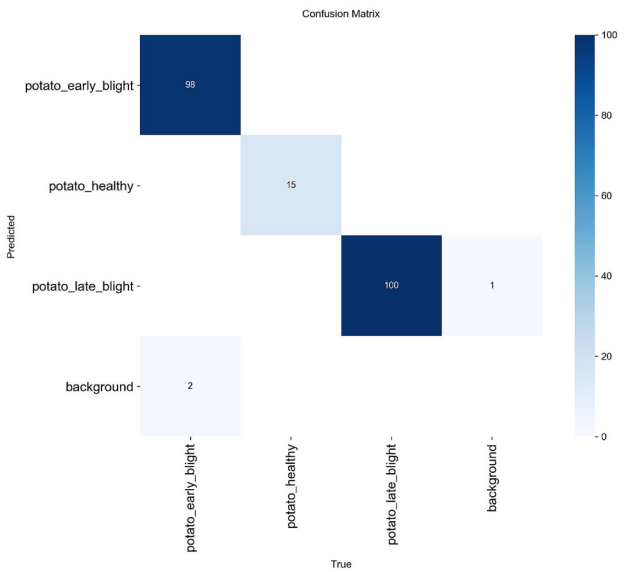


Fig. 6 Confusion matrix for NanoTeacher-YOLO26 on the held-out test split. The three leaf-level classes are well separated under the converted YOLO-format protocol

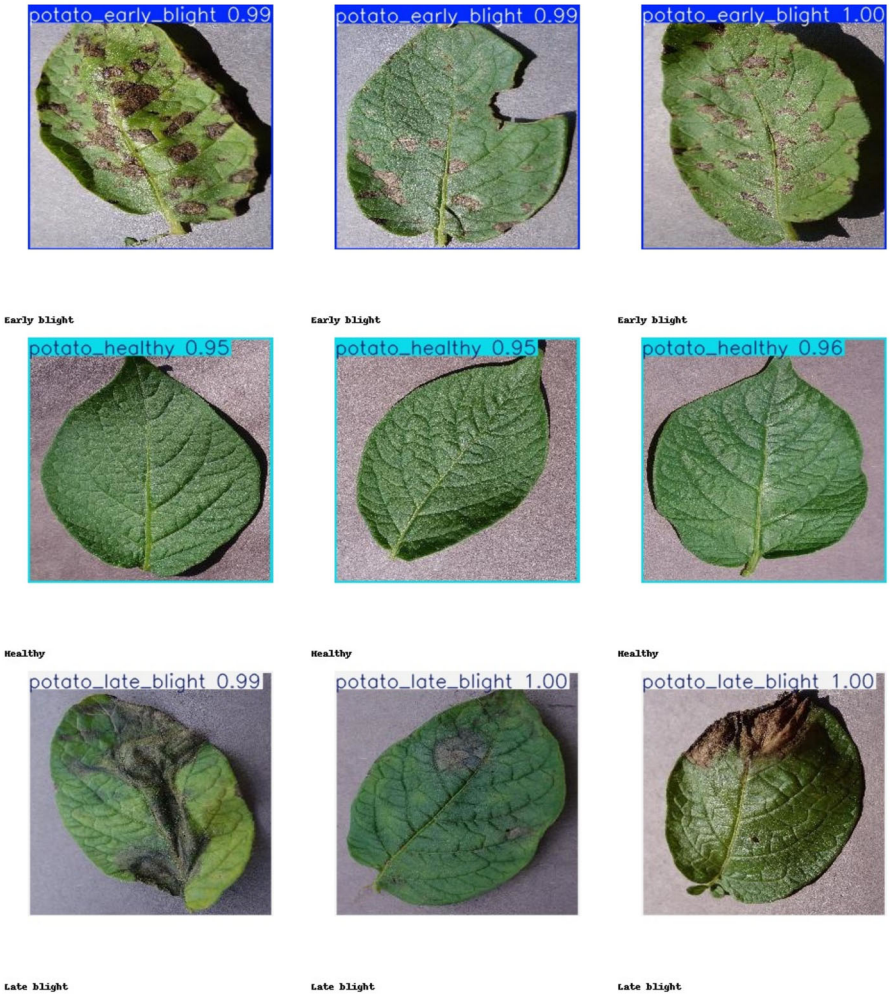


Fig. 7 Representative NanoTeacher-YOLO26 detection overlays on the test split. The deployed student produces class labels, confidence scores, and leaf-level boxes

detection protocol. The figures are presented separately to improve label readability in the revised manuscript.

The fast convergence is consistent with the role of the teacher. Instead of forcing the student to discover all spatial cues from hard labels alone, the teacher supplies intermediate supervision at multiple feature levels. The student therefore receives a richer learning signal early in training, which is useful when only a short training budget is allowed.

Detection Visualization

Representative detection results are shown in Fig. 7. The model predicts leaf-level boxes with high confidence and assigns the expected categories to early blight, healthy,

and late blight examples. Since the benchmark is converted from classification images, the goal is not pixel-accurate lesion segmentation. The correct interpretation is leaf-level disease detection with visible localization.

The detection overlays are important because they expose a limitation that a pure classification table cannot show. A detector that reaches high accuracy but localizes outside the leaf would be unsuitable for downstream screening. In contrast, the examples in Fig. 7 show that the compact student preserves the detector interface expected by field monitoring systems.

Teacher-Student Visual Alignment

Figures 8, 9, and 10 present attention heatmaps and teacher-student alignment examples. The distilled student concentrates activation on leaf tissue and symptom-rich regions. The teacher and student responses are not identical, which is expected because their capacities differ, but the student follows the same broad evidence layout after distillation. The alignment and box-overlay visualizations are now shown as separate full-width figures so that labels and spatial evidence are easier to inspect.

These visualizations support the method design. The attention term encourages the student to activate on disease-relevant leaf regions, while the boundary term discourages over-smoothed responses that would erase local symptom transitions. This is especially useful in potato disease recognition because the visual difference between early blight and late blight can depend on lesion distribution and local texture rather

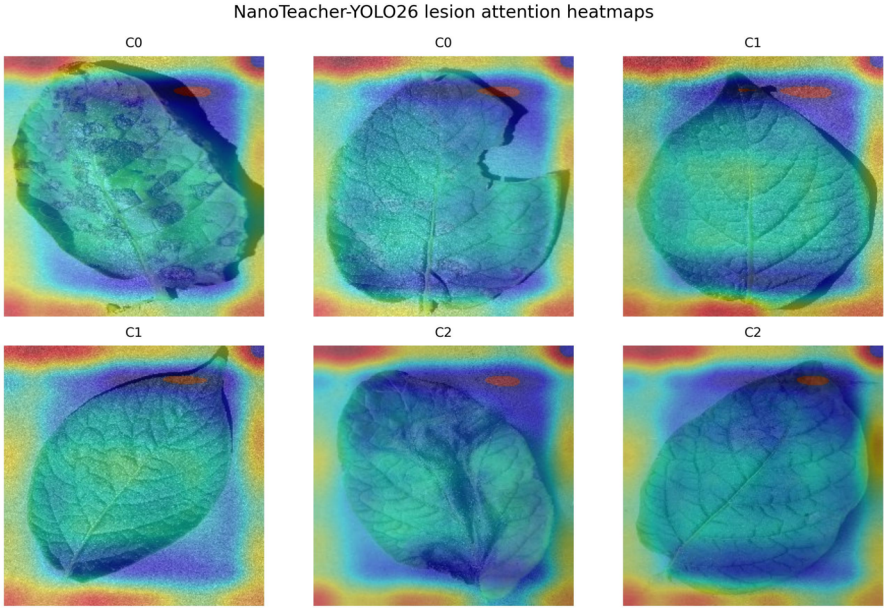


Fig. 8 Disease-evidence heatmaps generated from the trained NanoTeacher-YOLO26 student. Warmer areas indicate stronger spatial responses in the compact detector

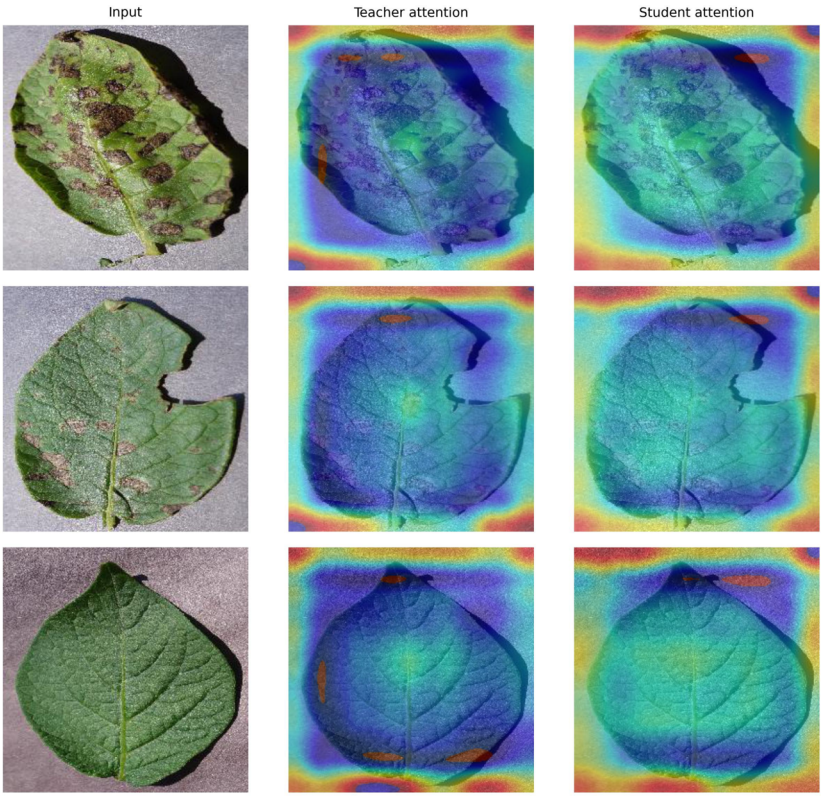


Fig. 9 Teacher-student attention comparison. The distilled student follows the teacher's broad disease-evidence layout while retaining a smaller YOLO26n backbone

than on global leaf silhouette alone. The visual evidence should be interpreted together with the saturated quantitative results: it explains how the same-footprint student changes its spatial behavior, even when rounded mAP values are identical across several nano baselines.



Fig. 10 Teacher-student detection box overlay. The deployed student produces leaf-level boxes similar to the teacher while preserving the YOLO26n deployment footprint

Table 5 Contextual comparison with representative potato disease recognition studies. Different rows are not necessarily evaluated under the same dataset or protocol

Study	Model or setting	Primary metric	Reported value
Hulsman et al. (2025)	EfficientNetV2 with activation-map localization for potato blackleg	Precision/recall	0.91/0.81
Alzakari et al. (2025)	Enhanced CNN-LSTM for potato disease detection	Accuracy	97.10%
El-Kenawy et al. (2025)	Metaheuristic-optimized CNN classifier	Accuracy	99.04%
Ozkan (2026)	ResNet50 features with DNN-3 classifier	Accuracy	98.56%
This work	Distilled YOLO26n detector	$mAP_{50}/mAP_{50:95}$	99.50/99.50%

Contextual Comparison with Potato Disease Studies

Table 5 positions the present work relative to representative potato disease studies. The comparison is contextual rather than a strict leaderboard because datasets, acquisition conditions, annotation protocols, and metrics differ. Many Potato Research studies report classification accuracy, while this work reports detector metrics on a leaf-level YOLO benchmark. The relevant distinction is that NanoTeacher-YOLO26 produces explicit detection outputs and explainable spatial visualizations while remaining a nano-scale model.

The contextual comparison reinforces the practical value of the proposed framework. High classification accuracy is already achievable in controlled potato leaf imagery, but deployment-oriented systems need more than a class label. NanoTeacher-YOLO26 provides a compact detector, explicit boxes, attention visualization, and teacher-student interpretability without increasing the inference-side model size.

Limitations

The study has three limitations. First, the dataset is derived from controlled PlantVillage images, so the reported performance should not be read as a complete field-robustness guarantee. Second, the converted labels are leaf-level boxes rather than manual lesion boxes or segmentation masks; therefore, the present benchmark evaluates leaf-level disease detection rather than lesion-level localization. Third, although the framework is designed to accept stronger foundation teachers, the present implementation uses a YOLO26s teacher to keep the experimental cost low and reproducible. Future work should extend the same distillation pipeline to real field images, multi-leaf scenes, manually annotated lesion boxes, and open-vocabulary teacher proposals from segmentation or grounding models.

Conclusion

This paper presented NanoTeacher-YOLO26, a foundation-to-edge distillation framework for lightweight potato disease detection. The method trains a deployable

YOLO26n student under the guidance of a stronger YOLO26s teacher and introduces a Disease-Aware Distillation Loss to transfer confidence-weighted feature geometry, spatial disease-evidence attention, and boundary-gradient priors. Unlike approaches that improve accuracy by enlarging the inference network, NanoTeacher-YOLO26 places the additional learning capacity entirely in the training stage. After optimization, the teacher and all auxiliary projection components are removed, allowing the final detector to retain the original parameter count and checkpoint size of YOLO26n.

Experiments on a YOLO-format PlantVillage potato benchmark demonstrate that NanoTeacher-YOLO26 reaches the saturated mAP regime of modern nano detectors while improving over the same-footprint YOLO26n baseline in both precision and recall. Because several baselines already reach the same rounded mAP ceiling, these results should be interpreted primarily as evidence of same-footprint precision–recall improvement, spatial alignment, and deployment value rather than as a large mAP gain. The proposed distillation strategy does not rely on a larger deployed architecture, but instead improves the representation quality of the compact student. In particular, the teacher-guided supervision helps the student learn more stable localization behavior and more discriminative spatial responses for early blight, healthy leaves, and late blight.

The visualization results further support the effectiveness of the proposed framework. Detection overlays show that the distilled student preserves reliable leaf-level localization, while attention maps and teacher-student alignment visualizations indicate that the compact model learns disease-relevant spatial evidence rather than relying only on global class-level shortcuts. The boundary-gradient component also provides a structured way to transfer local transition cues, which are important for recognizing symptom-related texture changes and disease-associated visual patterns in potato leaves.

Overall, NanoTeacher-YOLO26 provides a practical and reproducible route for converting teacher-side visual knowledge into an edge-ready agricultural detector. The study also has limitations: the current benchmark is derived from controlled PlantVillage images, and the annotations correspond to leaf-level boxes rather than manually delineated lesion regions. For field deployment, the trained YOLO26n student can be exported from the training framework to common edge formats such as ONNX, TensorRT, OpenVINO, or NCNN, followed by device-specific quantization when supported by the target hardware. Because the deployed checkpoint is only 5.12 MB and the inference graph contains no teacher, projection layer, attention branch, or boundary-gradient computation, it is suitable for evaluation on embedded GPUs or agricultural mobile platforms with camera input. A practical deployment pipeline would include image acquisition, lightweight preprocessing at 416-pixel resolution, on-device inference, confidence-threshold filtering, and periodic validation on field images to monitor domain shift. Future work will extend the framework to real field images, multi-leaf scenes, lesion-level annotations, and stronger foundation teachers such as open-vocabulary detection or segmentation models. These extensions may further improve the robustness and clinical-like interpretability of lightweight potato disease detection systems under complex agricultural environments.

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Data Availability The source images are publicly available from the PlantVillage leaf image release distributed through Mendeley Data as reported by Arun Pandian and Geetharamani (2019), DOI: https://doi.org/10.17632/tywbtsjrjv.1. The YOLO-format benchmark used in this study was constructed from the potato subset by assigning one leaf-level box to each image, as described in the "Dataset" section.

Declarations

Ethics Approval This article does not contain any studies with human participants or animals.

Conflict of Interest The authors declare no competing interests.

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